WALIFIT

Hybrid Athlete Operating System

Product Roadmap • V1 → V2.5

Product Identity & Positioning

waliFit is a Hybrid Athlete Operating System that combines AI coaching, workout building, nutrition intelligence, gamified consistency, and competitive social systems into one platform.

Core Loop: Train → Log → Progress → Compete → Repeat

Primary Target: Hybrid athletes — people balancing strength and conditioning, training for Hyrox-style events, or pursuing combined performance goals.

Secondary Target: Serious gym-goers who want accountability, structure, and something more engaging than a basic tracker.

> *Positioning note: waliFit's competitive moat is not any single feature — it is the integration of all of them under one identity, anchored by the Vitality Tree gamification system that no major competitor owns. Every individual feature (AI coaching, nutrition tracking, social leaderboards) exists elsewhere. The reason someone switches to waliFit is because nowhere else does training, nutrition, gamification, and social competition feel like one living system.*

Release Strategy

waliFit launches free to build a user base. Monetization is introduced in a later phase when new premium features justify it — not by gating core functionality that users already have.

Growth Phase (V1 – V2)	Monetization Phase (Post-V2 → V2.5)
100% free. Focus on user acquisition, retention, and product-market fit. No feature gates.	After V2 ships and the full core product is in users' hands, introduce a premium tier around new V2.5 features only — nothing existing ever gets gated.

V1 · LAUNCH

Training Core + Wali AI + Vitality Tree + WaliRun (GPS) + The Arena (Squads + Run Clubs) + Settings + Nutrition Logger

Goal of V1

Ship the complete core experience in one launch. Training, AI coaching, the Vitality Tree, and the social layer all go out together. Social accountability from day one means users have a reason to return even before a strong solo habit is formed — a streak your squad can see is far more powerful than one only you know about.

> *Why combine V1 and V1.5: Splitting social into a separate release would mean launching a training app with no audience. Streaks, PRs, and tree growth are all more meaningful when someone else can see them. The social layer does not add significant infrastructure risk — squads, a PR feed, and messaging are well-understood systems. Shipping them together creates network effects from the start.*

Feature 1: Wali AI (formerly 'Garden AI')

The central AI coaching experience. Renamed from 'Garden AI' to 'Wali AI' to make its purpose immediately clear to new users.

Program Architect

Generates custom training plans based on the user's goals, schedule, and equipment.

- Generate 4–6 week training blocks
- Support goals: strength, fat loss, muscle gain, hybrid performance, conditioning
- Adjust for available days, session length, and equipment
- Support beginner, intermediate, and advanced users

Daily Coach Chat

Conversational AI the user can talk to every day about training, recovery, and workouts.

- Answer training and general fitness questions
- Explain exercises and movement patterns
- Help modify today's workout in real time
- Provide substitutions for unavailable equipment
- Give recovery and general fitness guidance

Proactive Cold-Start Personality

Wali AI speaks first. The coaching relationship starts before the user ever asks a question.

– Triggered automatically when onboardingComplete flips to true
– Wali AI sends the opening message — it does not wait for the user
– Message is personalised to the user's stated goals from onboarding
– Example for a strength goal: 'Hey — I've looked at your goals and I've got some thoughts on where to start. What does your current week of training look like?'
– Example for a hybrid/Hyrox goal: 'Your conditioning base is going to be the foundation here. When was the last time you ran 5K and roughly how long did it take?'
– The opening question is always exactly one question — never multiple
– This one moment sets the tone for the entire coaching relationship
– Engineering cost: ~2 hours — a system prompt update and a triggered API call on onboarding completion
– Uses the coldStartCoach system prompt (see Technical Architecture Section 4.4)

Workout Adjustment Engine

Allows the AI to modify workouts on the fly without leaving the coaching experience.
– Shorten or expand workout volume
– Substitute equipment-dependent movements
– Switch between gym and home-friendly formats
– Works seamlessly inside Daily Coach Chat

Feature 2: Hybrid Performance Engine

The workout and training core — the primary input surface for the entire product.

Workout Logging

Users log and track completed workouts. This is the primary data source for gamification, streaks, and eventually social features.
– Log standard strength sets and reps
– Log time-based and interval-style conditioning workouts
– Add notes per session
– Save results to history
– Mark completion status

Custom Workout Builder

Advanced users want control over their programming — this gives them it.
– Build workouts from scratch
– Create weekly splits
– Set sets, reps, rest periods, or time intervals
– Mix strength and conditioning sessions
– Save, duplicate, and edit templates
– Assign plans to self

AI-Generated Plan Editing

Users should not be locked into AI output — they can modify and combine it with their own judgment.

- Edit exercises, sets, and reps in AI-generated plans
- Replace exercises or ask AI to revise a section
- Combine manual programming with AI support

Basic Exercise Library

A searchable reference catalog.

- Search by name, category, or muscle group
- Movement type classification
- Basic descriptions and targeted muscles
- No video library required in V1 — a clean searchable catalog is sufficient

Plate Math Calculator

A utility for compound lift setup — fits naturally with serious strength users.

- Enter target weight, get plate distribution
- Visual bar-loading guide
- Support standard barbell formats

Interval & Conditioning Support

Supports hybrid athletes who train with timed and rounds-based conditioning.

- Timed work/rest formats
- Rounds-based conditioning
- Basic HIIT logic
- Conditioning logs in workout history
- Keep practical and simple in V1 — specialization comes in V2.5

Feature 3: Vitality Garden

The gamification core of waliFit and the product's primary identity differentiator. This is what makes waliFit feel unlike any other fitness app.

Vitality Tree

A living digital tree on the home screen that reflects the user's consistency and daily habits. This is not decorative — it is the primary reward surface.

- Grows visually based on the user's daily health score
- Acts as the main retention and motivation mechanism

– Central presence on the home screen

Growth Inputs & Health Score

The tree responds to daily inputs — but workout completion is only active on days the user has scheduled as training days.

– Hydration: 30% of daily health score — applies every day
– Protein intake: 30% of daily health score — applies every day
– Workout completion: 40% — only active on user-scheduled training days
– On rest days, the activity pillar is neutral — the tree does not penalize planned recovery
– Rest is treated as part of training, not a failure state
– Optional: log active recovery, mobility, or a walk on rest days as a bonus input — never required

Training Day vs Rest Day Logic

The tree understands the difference between a missed workout and a planned rest day.

– User sets their weekly training schedule during onboarding (e.g. Mon / Wed / Fri)
– Training days: workout completion tracked and weighted at 40%
– Rest days: activity pillar is inactive — hydration and protein still apply
– Missing a training day hurts the tree; skipping a rest day workout does nothing
– Schedule can be adjusted week to week via the Calendar
– AI-generated plans automatically populate the training day schedule

Tree Growth States

Five visual stages that make progress feel tangible.

– Wilted — missed multiple days
– Recovering — inconsistent
– Sprout — building habits
– Growing — consistent progress
– Full Vitality — daily goals consistently hit

Wilt Mechanic

The tree declines when the user is inconsistent — this makes the system feel meaningful rather than decorative.

– Tree health drops when goals are missed for multiple days
– Streak is put at risk
– Visual feedback reinforces the value of consistency

Consistency Streak

A major retention feature tied directly to the tree.

– Daily streak count
– Milestone recognition at key thresholds
– Streak tied to tree vitality state
– Social visibility unlocked in V1.5

Feature 4: waliFit Calendar

The central time layer of the product. Everything in waliFit — workouts, nutrition goals, streaks, challenges, rest days, and AI plans — is anchored to real dates. The calendar is what makes the product feel like a living schedule rather than a disconnected set of tools.

Daily View

What does today look like? The user's single most important screen.

– Scheduled workout for today (from AI plan or custom plan)
– Whether today is a training day or rest day
– Nutrition targets for the day
– Hydration goal progress
– Vitality Tree status and today's health score
– Any active squad challenges or weekly objectives

Weekly View

A full-week overview that shows the training/rest rhythm at a glance.

– Training days and rest days clearly marked
– Completed vs upcoming workouts
– Streak continuity visible across the week
– Nutrition consistency indicators per day
– Tap any day to see its full detail view

Monthly View

Zoom out to see patterns, consistency, and upcoming milestones.

– Color-coded day states: completed, rest, missed, upcoming
– Streak runs visible at a glance
– Upcoming squad challenges and weekly challenge deadlines
– PR and milestone history browsable by date

Time-Based Linking

Every feature in waliFit connects to a specific date — nothing floats without context.

– AI-generated plans are mapped to real calendar dates on creation
– Workout logs are timestamped and retrievable by date

– Nutrition entries are tagged to the day they were logged
– Streak logic is date-driven — the system knows what day it is and what was expected
– Squad challenge deadlines show as calendar events
– PR feed events are linked to the date they occurred
– Rest days set by the user are blocked on the calendar so the tree never penalizes them

Feature 5: Onboarding & User Profile

Collects the information needed to personalise Wali AI plans, nutrition, and the overall experience. Must be completable in under 3 minutes — every extra screen costs sign-up rate.

Goal & Training Setup

Sets the foundation for all AI personalisation.

– Primary and secondary goal selection (strength, conditioning, fat loss, muscle gain, hybrid)
– Training experience level
– Available workout days and session length
– Equipment availability
– Optional: body weight, dietary pattern, injuries or limitations

Units Preference — kg or lbs

Must be set in onboarding before any weight data is stored. Changing this after launch requires a live data migration.

– User selects kg or lbs during onboarding — shown on the first screen after sign-up
– All weight values displayed in the user's chosen unit throughout the app
– Database always stores in kg — UI layer converts on display
– Plate calculator adapts to the selected unit system
– Can be changed in Settings at any time — triggers a re-render of all stored values
– Default: detect from device locale (US → lbs, everywhere else → kg)

Daily Targets Setup

Sets the baseline numbers the Vitality Tree scores against each day.

– Daily protein target in grams (Wali AI suggests based on body weight and goals)
– Daily water target in litres or fl oz based on units preference
– Both can be adjusted at any time in Settings
– Wali AI generates a personalised suggestion — user can override

Onboarding Skip & Return Path

A user who closes the app mid-onboarding must return to the exact step they left — not a blank screen. This is a real conversion issue if not handled.

– onboardingComplete and onboardingStep fields on User model track progress
– Each step saves to the server immediately — not on final submit
– On app reopen: if onboardingComplete is false → navigate directly to the incomplete step
– Users can skip optional steps (physical details, injury notes) and continue
– Onboarding is completable in under 3 minutes on the happy path
– Minimum viable onboarding: goals + units preference — everything else is optional

Multimodal Profile Import

Upload a screenshot of your old fitness app. Wali AI reads it and pre-fills your entire profile. Solves cold start in under 60 seconds.

– Optional step shown on the first onboarding screen: 'Already training somewhere? Import your profile'
– User uploads a screenshot from Hevy, MyFitnessPal, Strava, Strong, or any fitness app
– Gemini Vision API reads the screenshot and extracts: current PRs, body weight, training frequency, goals, recent workouts
– Extracted data pre-fills all onboarding fields — user reviews and edits before confirming
– Fields that cannot be extracted are left blank for manual entry
– Engineering cost: ~1 day — one Gemini Vision API call, a parsing layer, and a pre-fill UI step
– Uses the onboarding_import task type routed to Gemini Vision (see Technical Architecture Section 4.2)
– Completely optional — skip button always visible, no friction if user prefers manual entry

First-Time User Experience

A new user with zero data must feel welcomed, not overwhelmed. The cold start problem is real — an empty app feels broken.

– After onboarding, Wali AI immediately offers to generate a first training plan
– Squad discovery prompt shown on first open of the Arena tab — not an empty screen
– The Vitality Tree shows a seed/sprout state on day one — not a dead tree
– Today's workout appears on the home screen immediately after plan generation
– First workout completion triggers a milestone celebration — make the first win feel big
– Prompt to invite one friend shown after first workout logged — capitalises on early motivation

Feature 6: Simple Nutrition Logger

This feature must be in V1, not V2. The Vitality Tree scores protein intake at 30% of the daily health score — without a way to log protein, one third of the tree is permanently broken at launch. This is not Smart Fuel. It is the minimum needed to make the tree work.

> *Why this is V1 and not V2: Smart Fuel (full nutrition with barcode scanning, food rating, AI meal parsing) is correctly placed in V2. But the tree's protein pillar needs input data from day one. A simple*

> *daily protein total input costs two days of engineering and prevents a fundamental product failure at launch.*

Daily Protein Log

One-tap daily protein entry — no food database, no scanning, just a number.

- User enters total protein grams consumed for the day
- Can log incrementally throughout the day (add 30g, then 40g later)
- Shows progress toward daily target as a simple bar
- Feeds directly into the tree's protein pillar score
- Accessible from the home screen and daily calendar view with one tap
- In V2 this input is replaced by Smart Fuel's tracked totals — no migration needed

Daily Hydration Log

Water intake tracking — the other undefined tree input.

- User logs water in glasses (default) or ml/fl oz based on units preference
- One-tap glass addition on the home screen — frictionless is critical for daily use
- Progress ring or bar showing intake vs daily target
- Feeds directly into the tree's hydration pillar score (30% of daily score)
- Optional: set a reminder schedule inside notification settings

Feature 7: Settings Screen

Not mentioned anywhere in the original spec but required for launch. Every preference, account action, and legal link lives here.

Profile & Preferences

User-configurable settings that affect how the app behaves.

- Display name and profile photo
- Units preference — kg or lbs (with instant UI update)
- Daily protein target — overrides Wali AI suggestion
- Daily water target
- Training day schedule — overrides onboarding selection
- Default rest timer duration
- Language preference (future use)

Notification Settings

Per-type notification controls — respects users who granted permission but want fewer pings.

- Workout reminders — on/off, set time
- Hydration nudges — on/off, set frequency
- Streak at-risk alerts — on/off

– Squad activity alerts — on/off
– PR and milestone celebrations — on/off
– In-app fallback summary card shown when all notifications are off

Account Management

Apple App Store requires account deletion to be available in-app — this is a hard requirement.

– Change email address
– Change password
– Connected accounts (Apple Sign-In, Google Sign-In)
– Export my data — downloadable JSON of all workout logs, nutrition, and settings
– Delete my account — requires confirmation, deletes all data within 30 days, sends confirmation email
– Account deletion is permanent and disclosed clearly before confirmation

Legal & Privacy

Required links for App Store compliance.

– Privacy Policy — links to hosted Privacy Policy page
– Terms of Service — links to hosted Terms of Service page
– AI Disclaimer — one paragraph explaining Wali AI is a fitness guidance tool, not a medical professional
– Data collection summary — plain-language explanation of what is collected and why
– Manage data sharing preferences — opt out of AI training data collection

AI Processing Disclosure

Required for 2026 App Store compliance. Shows users exactly what happens on-device vs in the cloud — in plain language, not legal text.

– Displayed as a clean two-column list inside Settings → Privacy
– On-device (never leaves your phone): GPS tracking during runs, rest timer, vitality tree display, streak counter, exercise library, offline workout logging, MMKV local storage
– Sent to Anthropic's servers: messages you send to Wali AI, your training profile context (goals, recent workouts, plan), nutrition averages
– Sent to Google's servers: nutrition text parsing, food photo analysis, simple coaching questions
– Sent to waliFit servers: workout logs, run data, nutrition logs, squad activity, PR records
– Link to full Privacy Policy for users who want the complete picture
– Engineering cost: ~3 hours — a static Settings screen section with clear iconography
– Note: Apple's 2026 App Store privacy requirements increasingly demand in-app disclosure beyond the nutrition label — this proactively satisfies that requirement

Feature 8: The Arena

The competitive social heart of waliFit. The Arena is where squads form, run clubs compete, PRs get celebrated, and accountability happens. It has two flavours — Workout Squads and Run Clubs — but one infrastructure. Both live inside The Arena, both share the same leaderboard and messaging systems, and both contribute to the squad forest and PR feed. A user can be in both.

> *Naming note: the in-product naming is 'The Arena' for the overall competitive layer. Internally 'Squads' refers to workout-focused groups and 'Run Clubs' to run-focused groups — but both are squads under the hood, distinguished only by the squadType field. The V2.5 Global Arena competition system is the next tier of The Arena — platform-wide challenges that rise above individual squads.*

Workout Squads

Groups organised around strength and conditioning training — the original squad concept.

- Create a squad — name, join code, max 30 members in V1
- Leaderboard metrics: workouts completed, vitality score, streak length, PRs hit this week
- Squad forest: all member trees rendered together — a wilting tree is visible to everyone
- Group messaging for workout planning, check-ins, and accountability
- Shared programme support in V2 — run the same training block together
- Squad challenges in V2 — shared goals and collective targets

Run Clubs

Groups organised around running — same infrastructure as Workout Squads, different leaderboard focus.

- Create a Run Club — same join code mechanic, same 30-member cap
- Optional focus distance: '5K Club', '1 Mile Club', or 'All Distances'
- Run leaderboard: ranked by personal best per standard distance (1mi, 2mi, 3mi, 2K, 5K)
- Secondary metric: total km/miles logged this week per member
- Run challenges: club admin posts a distance, members submit times within a deadline
- Same squad forest, same group messaging, same PR feed as Workout Squads
- A user can be in both a Workout Squad and a Run Club simultaneously

The Arena Leaderboard

Two tabs inside every squad — one for training, one for running. Always present, always live.

- Training tab: workouts, vitality score, streaks, total volume
- Run tab: best times per standard distance, weekly mileage — shown in all squads not just Run Clubs
- A Workout Squad member who also runs still sees their run times on the Run tab
- A Run Club member who also lifts still sees their workouts on the Training tab
- This cross-pollination is intentional — waliFit is hybrid, not siloed

PR Feed

An activity feed for fitness milestones — automatically generated, never manually posted.

- Strength PR: '100kg squat — new all-time best'
- Run PR: '5K: 24:31 — 45 seconds faster than previous best'
- Workout milestone: 'Completed 4 workouts this week'
- Streak milestone: '30-day streak — longest ever'
- Tree growth: 'Vitality Tree reached Full Vitality'
- Simple engagement: Cheer and Nudge reactions

Friend System & Messaging

Controls who can message whom — prevents spam, keeps conversations meaningful.

- Friend request states: None → Pending → Friends → Blocked
- Direct messaging unlocked only after mutual friendship acceptance
- Group messaging inside every squad and run club
- All messaging is text-only in V1 — lightweight and offline-compatible

Feature 9: Rest Timer

One of the most-used features in any serious gym app. Used every single session during strength training — if it is missing, users notice immediately.

In-Set Rest Timer

A simple, always-accessible countdown timer that lives inside the active workout screen.

- Start timer after logging a set with one tap
- Configurable rest duration per exercise (default can be set in preferences)
- Audible and haptic alert when rest period ends
- Auto-suggests next set when timer completes
- Timer continues running if user navigates away from the screen
- Works fully offline

Feature 10: Smart Notifications

Notifications are a primary retention mechanism — without them the app goes silent and users forget to open it. These should feel helpful and timely, never spammy. Permission timing is critical on iOS — ask too early and users deny it permanently.

> *Push permission timing: Do NOT request notification permission at app launch or during onboarding. The right trigger is immediately after the user completes their first workout. At that moment the user has experienced the product, understands the value, and is in a positive emotional state. Asking at launch — before any value is delivered — results in a high denial rate that permanently kills your best retention mechanism.*

Workout Reminders

Reminds users on their scheduled training days.

- Fires on calendar-scheduled training days at a user-set time
- Message references today's planned workout by name
- Skipped on rest days automatically
- Fully customisable — time, days, and on/off per type

Hydration Nudges

Mid-day reminders to log water intake and hit the daily hydration goal.

- Periodic nudges spaced across the day
- Pauses after the daily hydration goal is already met
- Frequency configurable in settings

Streak & Tree At-Risk Alerts

Notifies the user before they lose a streak or let the tree wilt.

- Fires in the evening if the daily goal has not been completed
- Warns when tree is at risk of wilting due to consecutive missed days
- Streak milestone celebrations (7 days, 30 days, 100 days etc.)

Squad Activity Alerts

Keeps users connected to their squad without requiring them to manually check.

- Notify when a squad member posts a PR or hits a milestone
- Notify when someone reacts to your PR feed event
- Notify when a squad member sends a direct message
- All squad notification types individually toggleable

Feature 11: Offline Mode

Gym WiFi is notoriously unreliable. The app must work without a connection at the exact moment it is needed most — during a workout.

Offline Core Features

The essential surfaces that must function without any internet connection.

- Active workout screen — log sets, reps, weights, and notes
- Rest timer — fully local, no network dependency
- Exercise library — cached locally on first load
- Today's workout from the active plan — cached on open
- Vitality Tree and streak display — reads from local state
- Calendar view — reads from locally cached schedule

Sync on Reconnect

All offline activity syncs cleanly when connection is restored.

- Workout logs upload automatically on reconnect
- Nutrition entries queued offline and synced on reconnect
- Streak and health score recalculated server-side after sync
- No data loss if the app is closed before reconnecting
- Sync status indicator visible to the user

Feature 12: WaliRun

Running is native to waliFit. Hybrid athletes run — Hyrox has a 1km run between every single station. WaliRun extends the existing Hybrid Performance Engine to treat run sessions as a first-class workout type. It does not compete with Strava. It makes running data part of the waliFit loop — feeding the Vitality Tree, posting PRs to the squad feed, and giving Wali AI the context to coach conditioning properly.

> *GPS decision: External validation feedback suggested deferring GPS to V1.5 as a scope concern. After review, the decision is to ship foreground GPS in V1 — screen stays on during the run, uses expo-location, approximately 5–7 days of engineering. This is not a large lift and manual-only entry makes WaliRun feel incomplete for a hybrid athlete product. Background GPS (screen-off) remains in V2.*

Run Logger — Foreground GPS (V1)

Live GPS tracking using the device's location. Screen stays on during the run. Covers the full WaliRun use case for every standard distance.

- Start a run with one tap — GPS acquires lock before countdown begins
- Live display: current pace, average pace, elapsed distance, elapsed time
- Audio cues at every km or mile split (based on units preference)
- Auto-stops and celebrates when target distance is reached on standard distance modes
- Free Run mode available — no target distance, stops manually
- Route stored as encoded polyline on WorkoutLog — displayed on the post-run stats screen
- Works with screen on — foreground only in V1. Background GPS (screen-off) ships in V2
- Offline compatible — GPS is on-device, route cached locally, syncs on reconnect
- Implementation: expo-location + Haversine distance formula, approximately 5–7 days of engineering

Standard Distance Modes

Preset race-standard distances — the primary competitive format in WaliRun.

- Available distances: 1 mile, 2 mile, 3 mile, 2K, 5K
- Both metric and imperial distances always available regardless of units preference
- Metric users see pace as min/km. Imperial users see min/mile. Both are always shown

– Auto-stop: app detects when target distance is reached via GPS and ends the session
– Finish time, pace, and route logged automatically — no manual entry
– Personal record detected automatically — triggers PR feed post if it is a new best

Individual Run Stats Report

A dedicated post-run summary screen that appears immediately after every run. This is why people open Strava straight after finishing — waliFit gives them the same moment.

– Distance and total time — the headline numbers
– Average pace per km and per mile — both always shown
– Fastest split and slowest split — shows where the run was strong and where it faded
– Splits breakdown — pace per km or mile shown as a simple bar chart
– Delta vs personal best for that distance: '+1:24 off your 5K PR' or 'NEW 5K BEST — 45s faster'
– Run type label: Easy / Tempo / Race / Interval — set before the run
– Route map showing the GPS track
– Shareable stats card — one tap generates a clean card showing distance, time, pace, and route thumbnail for sharing externally
– Estimated effort based on pace vs PR history (Easy / Moderate / Hard / Max)
– All stats generated client-side from data already in the WorkoutLog — no extra API calls

Run Personal Records

Automatic PR tracking per standard distance — uses the existing PRRecord schema unchanged.

– exerciseName: '5K Run', '1 Mile Run', '2K Run' etc.
– value: total seconds — unit: 'seconds'
– New fastest time triggers a PR exactly like a strength PR — PR feed post, squad notification, tree milestone
– All-time bests per distance shown on the Run PRs tab in the user profile
– PR history: every run for a distance plotted over time shows progression

Personal Run Stats (Solo Tracking)

For users who run purely for personal improvement — not squad competition. Every run builds a private performance history.

– Weekly run summary: total distance, total time, number of runs, average pace trend
– Distance progression: how far the user has run this week, month, year
– Pace trend per distance: is your 5K pace improving month over month?
– Streak integration: consecutive weeks with at least one run
– No squad required — solo runners get full stats without joining a Run Club
– In V2: these stats feed into the Performance Dashboard alongside strength trends

Run Personal Records

Automatic PR tracking per standard distance — uses the existing PRRecord schema with zero schema changes.

- exerciseName set to '5K Run', '1 Mile Run' etc. — same field used for squat PRs
- value stored in total seconds — unit set to 'seconds'
- New fastest time triggers a PR exactly like a strength PR — PR feed post, squad notification, tree milestone
- All-time best per distance shown in a dedicated Run PRs tab in the user profile
- Pace improvement visible: 'Your 5K improved by 45 seconds this month'

Run History

A full log of every run with route, stats, and splits — browsable, filterable, connected to the Calendar.

- All run sessions visible in workout history filtered by session type
- Each entry shows: distance, time, average pace, route map, date
- Tap any run to open the full Individual Run Stats Report for that session
- Filter by distance to see progression on a specific race distance over time
- Planned runs from Wali AI appear as calendar sessions on training days
- V2 adds: background GPS so screen can lock during runs

Feature 12a: Run Clubs

Run Clubs are Squads with a squadType field set to 'run_club'. No new infrastructure, no new tables, no new API routes beyond filtering. The Squad leaderboard gains a Run tab showing best times per distance and weekly km total.

> *This is the key architectural insight from the external validation: building a parallel RunClub system alongside Squads would mean duplicating messaging, member management, join codes, and leaderboard logic. Extending Squad with two fields delivers identical user-facing functionality at a fraction of the cost.*

Run Club Creation

Exactly like creating a Squad — with an optional running focus.

- Create a squad and set squadType to 'run_club'
- Optional runFocusDistance field: '1mi', '2mi', '3mi', '2k', '5k', or 'all'
- Run clubs appear in a separate Run Clubs tab alongside regular squads
- Hard cap: 30 members per squad/run club in V1 — prevents expensive leaderboard queries on large groups
- Cap increases as infrastructure scales — defined explicitly to protect database performance

Run Club Leaderboard

The Squad leaderboard gains a Run tab — no new leaderboard system needed.

– Run tab shows best times per standard distance for all club members
– SQL query: SELECT MIN(value) FROM PRRecord WHERE exerciseName IN run distances AND userId IN club members
– Ranked ascending by time — fastest at top
– Weekly km total per member shown as a secondary metric
– New run PR automatically updates club position — no manual refresh needed

Run Stats Sharing

Runs feed the same social layer as strength sessions.

– Completing a standard distance run auto-posts to the PR feed
– PR card shows distance, time, pace, and improvement vs previous best
– Run club members see the PR in their squad feed
– One-tap shareable stats card via native share sheet — shows distance, time, pace on a clean card
– No separate 'run social' system — the PR feed handles everything

Feature 12b: WaliRun × waliFit Integration

Every run connects to the systems already built. This is what makes waliFit feel like one coherent product — not a collection of features bolted together.

waliFit Feature	How WaliRun connects
Vitality Tree	Run on a scheduled training day counts as full activity pillar (40% of daily score) — same as any strength session
Calendar	Wali AI schedules run sessions as conditioning days. Planned run shows on today's home screen as 'Easy 5K — scheduled'
Wali AI	Last 5 run logs and run PRs injected into every AI context call. Wali AI says 'your 5K pace suggests Z2 base is ready — add tempo work next block'. run_coaching AI task routes to Gemini Flash (cheap, fast, sufficient for pace analysis)
Streak System	Completing a scheduled run counts toward streak — same logic as any workout type
PR Feed	Run PRs post alongside strength PRs. Squad members see '5K: 24:31' next to '100kg squat'. Same feed, same reactions
Smart Notifications	Run reminder on scheduled run days. PR celebration when a distance PR is broken
Offline Mode	Manual entry works fully offline — no GPS, no network. Syncs on reconnect
Progress & Analytics (V2)	Pace trend charts per distance over time. Weekly run volume. PR timeline per distance

Station-to-Run Timer (V2.5)	Compromised Run mode uses GPS (added in V2) to measure run pace immediately post-station — waliFit's unique hybrid event metric
Wearable Integration (V2.5)	Heart rate during GPS runs from Apple Watch or Garmin — average HR per km in run history

V2 · NUTRITION + INTELLIGENCE DEPTH

Smart Fuel, Tree Biomes, Squad Forests, RPE Timer & Advanced Gamification

Goal of V2

Add nutrition intelligence and deepen the gamification and social systems now that the core loop and social layer are both validated. V2 builds on what already exists rather than introducing entirely new infrastructure.

Feature 13: Smart Fuel

The nutrition intelligence system. Launching in V2 (not V1) allows the team to focus on getting training and social right first, and prevents a half-built nutrition experience from creating a bad first impression.

Text-Based Meal Input

The simplest nutrition entry path — users describe their meal in plain text.

- User types something like: 'chicken, rice, broccoli'
- AI parses the description and estimates calories, protein, carbs, and fats
- No database lookup required — estimation is sufficient for the use case

Product Barcode Scanner

Retrieves accurate nutrition data for packaged foods via external API.

- Scan barcode using device camera
- Fetch nutrition data from food database API
- Return calories, protein, carbs, fats, and ingredients
- Recommended APIs: Open Food Facts (free, open-source) or Nutritionix (higher accuracy, freemium)
- Do not build an internal food database — use an established API

Food Performance Rating

A differentiated nutrition experience — simple traffic-light scoring rather than raw macros.

– Rating tiers: Green (performance-positive), Yellow (neutral/moderate), Red (poor choice for goals)
– Scoring factors: protein quality, sugar content, calorie density, ingredient quality, sodium, goal relevance
– Define scoring formula explicitly before building — inconsistent AI scoring will damage trust
– Suggested formula: Protein score (25%) + Sugar penalty (25%) + Ingredient quality (25%) + Goal alignment (25%)

Ingredient Analysis

Light ingredient guidance for packaged foods — stay lightweight, avoid overpromising.

– Flag clearly poor-quality ingredients (hydrogenated oils, excessive additives)
– Identify high-sugar or heavily processed profiles
– Simple warning signals relevant to athletic performance
– Explicitly labeled as guidance, not clinical nutrition advice

Feature 14: Progress & Analytics

All that logged data needs somewhere to live visually. This turns workout history into insight and makes the effort of logging feel worthwhile.

Personal PR Vault

A personal record board for every tracked lift and movement.

– All-time PR per exercise automatically tracked from logs
– PR history — see when each record was set
– Browsable by movement category or date
– PR achievements feed into the social PR feed automatically

Strength Trend Charts

Visual progress tracking per lift over time.

– Line chart of best set weight per exercise over weeks and months
– Volume trends — total sets and reps per session
– Filter by exercise, date range, or training block
– Clearly shows plateaus, peaks, and progression

Workout Frequency & Consistency Stats

Shows how consistently the user is actually training.

– Workouts completed per week over time
– Training day adherence rate — scheduled vs completed
– Average session length
– Rest day vs training day balance

Feature 15: Body Metrics Tracking

Ties directly into goal tracking and nutrition — many users care about physical changes as much as performance numbers.

Weight & Measurement Logging

Simple manual input for body metrics over time.

- Log body weight with date
- Optional body measurements (waist, chest, arms, legs etc.)
- Weight trend chart over time
- Correlates with nutrition logs and training frequency

Progress Photos

Optional photo log for visual progress tracking.

- Upload photos tagged to a specific date
- Stored privately — never shared without explicit action
- Side-by-side comparison view between any two dates
- Clearly separated from the social feed

Feature 16: Advanced Gamification

Deepens the Vitality Garden system to increase retention and reward variety.

Weekly Challenges

Rotating gamified objectives to keep the daily experience fresh.

- Example: Hit hydration goal 5 days this week
- Example: Complete 4 workouts
- Example: Hit protein target 3 days in a row
- Rewards feed into the Vitality Tree and streak systems

Fitness Bingo

A randomized challenge card system that creates variety in the habit loop.

- Weekly or monthly bingo boards
- Complete squares by finishing fitness or nutrition tasks
- Earn in-app progression rewards on completion

Reward Unlocks

Cosmetic and identity-based progression items — creates a sense of earning without monetary prizes.

- Themes and color schemes
- Avatar items
- Tree skins
- Profile badges
- AI voice or personality options

Nexus Seed Economy

A symbolic reward item that connects challenges to the core Vitality Garden identity.

- Earn seeds through challenges or consistency milestones
- Use them to heal or accelerate tree progression
- Reinforces the garden identity and gives challenges a tangible reward

Feature 17: Tree Biomes

The type of tree that grows on the user's home screen changes based on their primary training goal. Costs almost nothing to implement but makes goal selection feel consequential and gives users a visual identity to own.

Goal-Linked Tree Types

Each training goal maps to a distinct tree species with its own visual identity.

- Strength goal → Oak tree — sturdy, wide, powerful growth stages
- Conditioning / endurance goal → Willow tree — tall, flowing, flexible
- Hybrid performance goal → Bamboo — fast-growing, resilient, multi-stemmed
- Fat loss / body recomposition goal → Cherry Blossom — seasonal, transformation-focused
- Muscle gain goal → Redwood — slow but massive, scale over time

Biome Switching

Users can change their goal and watch their tree species evolve.

- Changing primary goal triggers a tree transition animation
- Tree health state carries over — switching goal does not reset progress
- Biome choice visible on public profile and squad forest
- Seasonal visual variations possible as future cosmetic updates

Feature 18: Squad Forests

A shared visual space where all squad members' trees appear together. This is the most powerful accountability mechanic in the product — a wilting tree does not just affect the individual, it visibly affects the whole squad's forest.

Shared Forest View

All squad members' trees rendered together in a single visual environment.

- Each tree in the forest reflects its owner's current vitality state
- Forest overall health is an average of all member trees
- Wilted trees are visible to the whole squad — passive social accountability
- Full vitality trees make the forest lush and animated
- Accessible from the squad screen with one tap

Forest Health Score

A squad-level metric derived from the collective vitality of all members.

- Squad forest health score shown on squad leaderboard
- Squads can compare forest health in V2.5 Global Arena social filters
- Weekly forest health trend visible to all squad members
- Members whose trees are wilting receive a gentle nudge from the squad system

Feature 19: Smart Rest Timer Upgrades

Builds on the V1 rest timer with two intelligence layers — RPE-linked recovery and focus protection. Both are opt-in and clearly labelled.

RPE-Linked Rest Extension

When a user logs a high effort set, the AI automatically suggests extending their rest period.

- User logs RPE 8–10 after a set → AI suggests adding 30–60 seconds to rest timer
- Suggestion appears as a gentle prompt — user can accept or dismiss
- Helps prevent technical failure on subsequent sets
- Particularly valuable for heavy compound lifts in the Hybrid Performance Engine
- RPE logging added to the set completion screen in V2

Focus Alert Mode

Detects when a rest period is turning into a scrolling session and sends a haptic reminder.

- Opt-in feature — off by default, enabled in settings
- Detects phone inactivity during rest and fires a haptic alert if rest runs significantly over
- Designed for users who identify as students, founders, or easily distracted
- Alert fires only after rest period exceeds set duration by a configurable threshold
- Never interrupts an intentionally extended rest — only fires on forgotten timers

Feature 20: Squad & Social Expansion

Squad Challenges

Squads compete together rather than just observing individual rankings.

- Shared goals and collective streak targets
- Group challenge completion tracking
- Internal competition or collaborative modes

Shared Programs

Lets users and squads run the same training block together.

- Share custom plans with friends
- Share plans with full squad
- Run the same training block simultaneously

Expanded Feed Interactions

Makes the PR feed more conversational without losing its fitness focus.

- Comments and threaded replies
- Richer reaction options
- Remains secondary to accountability — not a general social feed

Feature 21: Home Screen Widget

A lightweight widget that keeps waliFit visible without requiring the user to open the app — high visibility, low effort to build, strong retention impact.

Widget Display

Shows the most motivating daily information at a glance.

- Current streak count
- Today's scheduled workout name
- Vitality Tree status (current growth stage)
- Whether today is a training day or rest day
- Tap to open directly into today's workout view

Feature 22: Referral & Invite System

Squads grow through word of mouth — a structured invite mechanic accelerates this without any ad spend.

User Referral Flow

A simple system that rewards users for bringing friends into waliFit.

– Each user gets a unique invite link
– Shareable directly to messages, social media, or squad chats
– Referral tracked when a new user signs up via the link
– Both referrer and new user receive an in-app reward (e.g. Nexus Seeds, badge)
– Referral stats visible in user profile

Squad Invite Shortcut

Makes growing a squad frictionless.

– One-tap share of squad join code or direct invite link
– Works inside squad group chat and externally
– Shows pending invites sent by the user

Feature 23: Female Training Considerations

Menstrual cycle phase significantly affects training adaptation, recovery capacity, and energy levels. A large portion of waliFit's audience is female. No major fitness app has addressed this well. Even a basic implementation in V2 would be a genuine differentiator — and better for the users it serves.

> *Why V2 and not later: This is not a nice-to-have. It is a product quality issue. Wali AI currently gives identical recovery and intensity recommendations regardless of physiological context. For female users this means the AI is systematically less useful. V2 is the right time — after the core loop is validated but before the AI coaching is considered complete.*

Cycle Phase Tracking

Optional input that gives Wali AI the context to make better recommendations for female users.

– Optional cycle phase input in user profile — never required, never surfaced unless enabled
– User logs approximate cycle start date or enables tracking mode
– Four phases detected: Menstrual, Follicular, Ovulation, Luteal
– Strictly opt-in — users who do not enable it see no change in the product

Phase-Aware Wali AI Coaching

Wali AI adjusts tone and recommendations based on cycle phase when tracking is enabled.

– Follicular and Ovulatory phases: higher intensity training appropriate — AI can push harder
– Luteal phase: recovery emphasis, reduced intensity suggestions, nutrition support for cravings
– Menstrual phase: AI recommends lighter sessions, mobility work, or rest without judgment
– Injected into Wali AI context alongside workout history — AI uses it naturally in coaching responses
– Wali AI never lectures or over-emphasises cycle phase — it uses it as quiet context, not a constant topic

Vitality Tree Adjustment

Rest days during high-impact cycle phases do not penalise the tree.

- User can mark a day as a cycle rest day — activity pillar treated as neutral regardless of training day status
- Prevents the streak and tree from punishing users for physiologically appropriate rest
- Optional — only available when cycle tracking is enabled

Feature 24: Performance Dashboard

The unified intelligence layer that connects training, nutrition, and recovery data into one view. No fitness app has done cross-system correlation well. This is waliFit's chance to show users something genuinely new — that their data means more when it talks to itself.

Performance Correlation Panel

Shows relationships between nutrition and strength performance — the insight users actually care about.

- Strength trend per lift overlaid with average weekly protein intake
- Did your squat go up on weeks where protein hit target? Now you can see it
- Workout completion rate vs average daily calorie intake
- RPE trend vs nutrition deficit — shows why undereating hurts training
- Correlation scores updated weekly from logged data

Weekly Report Card

Generated by Wali AI every Monday — a short paragraph summarising last week and one specific recommendation for the coming week.

- Workouts completed vs planned (with % adherence)
- Nutrition averages for the week — calories, protein, hydration
- Best PR hit during the week
- Streak status and tree vitality stage
- One specific, personalised recommendation for the coming week
- Delivered as a push notification with a tap-to-read summary in the app
- Major retention driver — gives users a reason to open the app every Monday

Consistency Heatmap

A GitHub-style contribution grid showing a full year of daily habit completion at a glance.

- Each day shows how many of the three pillars were hit (workout, protein, hydration)
- Colour intensity from empty → partial → full vitality
- A full year of this view is extremely motivating — users do not want to break the chain
- Tappable — tap any day to see its full detail log
- Visible on public profile so squads can see each other's annual consistency

Recovery Quality Score

A weekly composite score that tells users whether they are recovering or accumulating fatigue.

- Inputs: rest day adherence, sleep data (if wearable connected), workout completion, RPE trend
- Score displayed as a simple gauge: Undertrained / Optimal / Overreaching / Overtraining
- Wali AI flags when the score suggests a deload is needed
- Trend chart showing recovery quality over the past 8 weeks
- Wearable HRV data feeds this score automatically in V2.5

Body Composition Trend

Weight over time with training volume and nutrition overlaid — shows the full picture of recomposition.

- Body weight trend line with 7-day rolling average to smooth noise
- Training volume (total sets per week) overlaid on the same chart
- Average daily protein overlaid as a third line
- Wali AI flags meaningful patterns: weight dropping + strength up = good recomp, weight dropping + strength down = deficit too aggressive
- Progress photos browsable by date in the same view (V2 body metrics feature)

Squad Comparison Panel

Personal stats benchmarked against squad members — not just leaderboard rank, but full metric context.

- Your workouts this week vs squad average
- Your protein adherence vs squad average
- Your streak vs squad best and squad average
- Your vitality score rank within the squad
- Drives accountability without being aggressive — framed as context not competition

Feature 25: waliFit Commons

A global advice feed — structured like Reddit but purpose-built for hybrid athlete knowledge sharing. Not a general social feed. Every post has a category, every question gets a Wali AI response alongside community answers, and the best advice rises through a simple reputation system.

> *Why this is different from the PR feed: The PR feed is personal and squad-level — it shows your milestones to people who know you. waliFit Commons is global and anonymous-friendly — it is for asking questions you might not want to ask your squad, and learning from the wider waliFit community.*

Post Categories

Every post must have one category — this keeps the feed structured and searchable.

- Training Advice — programming questions, exercise technique, workout design
- Nutrition — meal planning, macro targets, food choices, supplementation
- Recovery — sleep, soreness, deload timing, injury management
- Hybrid Events — race prep, station strategies, transition training, event reviews
- Beginner Questions — no judgment zone for new users finding their feet
- Program Review — share your plan and get community + AI feedback
- Wins — personal victories that deserve more than a squad PR post

Wali AI Response

Every post automatically receives a Wali AI response — grounded in the RAG knowledge base, clearly labelled as AI.

- AI response generated asynchronously ~30 seconds after posting — not instant
- Pulls from exercise library, nutrition facts, and programming principles RAG
- Labelled clearly as 'Wali AI' with a distinct visual badge — never impersonates community
- Community can upvote or flag the AI response like any other comment
- If the AI response is flagged as wrong, it triggers a prompt review internally
- AI responses are free for all users — not gated behind Pro

Community Voting & Reputation

Simple system that surfaces the best advice without heavy moderation overhead.

- Upvote helpful comments — downvote unhelpful ones
- 'Helped me' tag — stronger signal than upvote, only shown when post author marks a comment
- Reputation score per user based on upvotes and 'Helped me' tags received
- High-reputation users get a visible badge on their profile and in squads
- Reputation is category-specific — a nutrition expert is not automatically a programming expert

Post Types

Four structured formats that keep Commons purposeful.

- Question — seeking advice, will receive AI + community responses
- Form Check — upload video or image of a lift for technique critique
- Program Review — share a training block for structured feedback
- Win — celebration post that also auto-generates a PR feed event in squads

Moderation System

Lightweight automated moderation that scales without a human team in the early stages.

- Wali AI pre-screens every post before it goes live — filters spam, harmful content, off-topic
- Posts flagged by AI go to a review queue — held, not deleted

– Community flagging available — three flags from different users triggers automatic review hold
– Clear community guidelines pinned at the top of every category
– Human moderation layer added once volume requires it — initially founder-moderated

V2.5 · GLOBAL COMPETITION + AI UPGRADE

Global Arena, Adaptive AI, Hybrid Event & Advanced Nutrition

Goal of V2.5

Introduce platform-wide competition and significantly upgrade the AI intelligence layer. These features require meaningful user volume to feel alive — they should not launch into an empty product. V2.5 is also the likely point where premium features are introduced.

> *Monetization launches here: V2 is fully free. Once V2 ships and the complete core product exists, V2.5 introduces a paid tier. Premium candidates: Global Arena access, advanced adaptive AI, Vision Meal Scanner, and cosmetic reward unlocks. Everything from V1 and V2 stays free permanently.*

Feature 26: Global Arena

A platform-wide competition system. This feature lives or dies on DAU — it needs users to feel competitive. Do not rush it.

Workout of the Week & Month

Official waliFit workouts released on a recurring schedule.
– Official workout card with defined standards and scoring
– Weekly cadence (lower stakes) and monthly cadence (higher prestige)
– Start and end dates with automatic leaderboard close
– Badge and reward opportunities tied to completion

Flexible Score Submission

Not all official workouts are time-based — the system supports multiple score types.
– Supported formats: time, reps, weight, distance, points
– RX and Scaled divisions for fairness
– Optional proof submission (image, video, or screenshot)
– Unverified vs. verified labeling — trust-based initially, review layer added later

Leaderboard Filters

Multiple filter layers make rankings feel personally relevant rather than overwhelming.

- Global leaderboard — worldwide rank
- Local filters — city, state, country
- Social filters — squad and friends
- User's personal rank always visible regardless of filter

Challenge Social Integration

Arena results feed back into the social system automatically.

- Auto-generated events: score submissions, rank achievements, squad placements
- Keeps the Arena connected to the core product loop
- No manual posting required from the user

Feature 27: AI Upgrade Layer

Upgrades Wali AI from assistance into genuine adaptive coaching. This tier is easier to build in V2.5 because it relies on the workout history database accumulated since V1.

Program Evolution Engine

AI adapts the training plan week-to-week based on actual user behavior.

- Detect missed workouts and adjust upcoming sessions
- Modify volume or difficulty based on performance trends
- Evolve programs week by week without user intervention

Recovery & Deload Suggestions

AI recognizes fatigue and responds proactively.

- Detect prolonged fatigue or declining performance patterns
- Recommend deload weeks when appropriate
- Suggest lower intensity sessions when needed

Improved Daily Coach Memory

Continuity across coaching conversations.

- Remember previous training plans and history
- Understand recent workout trends when answering questions
- More context-aware day-to-day adjustments

Feature 28: Advanced Smart Fuel

Vision Meal Scanner

Upgraded from the optional V1 feature into a reliable V2.5 experience.

- Upload or snap a meal image
- Meal recognition and ingredient estimation
- Calories and macro estimate
- Performance-oriented food feedback
- Clearly labeled as estimation — not medical nutrition advice

Energy & Bloat Tracking

Brings biofeedback into the nutrition system without becoming clinical.

- Tag meals with post-meal energy level
- Tag meals with bloating or digestive discomfort
- Surface patterns over time
- Help users identify possible food sensitivities

Feature 29: Conditioning Event Expansion

Hybrid Event Challenge Templates

Structured conditioning formats for hybrid event preparation.

- Station-based templates
- Conditioning circuits
- Predefined challenge modes
- Better support for users training for hybrid events

Improved Interval Builder

More flexible timed workout creation for serious conditioning athletes.

- Build fully custom interval structures
- Save interval templates for reuse
- Recurring timing patterns

Feature 30: Wearable Integration

Connects waliFit to Apple Watch, Garmin, and other devices to bring real physiological data into the coaching and recovery systems.

Heart Rate & Activity Sync

Pulls workout and health data from connected wearables.

– Heart rate data during conditioning sessions
– Step count on rest days feeds into activity bonus input
– Active calories logged against daily energy targets
– Compatible with Apple Health, Google Fit, and Garmin Connect

HRV & Recovery Integration

Feeds wearable recovery data into the AI coaching layer.

– HRV trends surface inside the Recovery & Deload Suggestions engine
– Low HRV days can trigger automatic intensity reduction suggestions
– Gives the AI real physiological signal rather than relying only on subjective feedback
– Premium feature — requires V2.5 AI Upgrade Layer

Apple Watch App

A lightweight companion app for the wrist.

– Start and log workouts directly from the watch
– Rest timer on wrist with haptic alert
– Streak and tree status glanceable from watch face
– Heart rate visible during active workout

V3 · MECHATRONICS & COMPUTER VISION

Bar Path Tracking, Velocity Monitoring & Advanced Form Analysis

Goal of V3

V3 is where your mechatronics background becomes a genuine technical moat. These features require computer vision infrastructure, real-time video analysis, and trained movement models — they cannot be rushed and should only be built once waliFit has the user base and revenue to justify the engineering investment. No consumer fitness app has done this well at scale. V3 is how waliFit becomes defensible.

> *Prerequisite for V3: Wearable integration (V2.5) should be fully shipped and stable before V3 begins. The sensor data pipeline and real-time data handling built for wearables provides the foundation for the more complex computer vision pipeline needed here.*

Feature 31: Bar Path Tracking

Uses the device camera to overlay a movement skeleton on the user during heavy compound lifts and track the path of the bar in real time.

Real-Time Bar Path Overlay

Visual feedback on bar trajectory during lifts.

- Camera detects and tracks the barbell using computer vision
- Overlays the bar's path as a line on the live video feed
- Ideal path reference shown alongside actual path for comparison
- Works for squat, deadlift, bench press, and overhead press
- Saves path data per session — visible in Analytics over time

Deviation Alerts

Flags significant bar path deviation in real time.

- Alerts when bar drifts outside an acceptable range from ideal path
- Post-set summary showing average deviation and worst rep
- Correlated with RPE and weight — helps identify technique breakdown under fatigue
- Builds a form history the AI can reference in coaching conversations

Feature 32: Velocity Loss Monitoring

Detects bar speed degradation across a set and suggests stopping before technical failure — a critical feature for the Hybrid Performance Engine's serious strength users.

Rep Velocity Detection

Measures bar speed on each rep using frame-rate analysis from the device camera.

- Calculates velocity of each rep within a set
- Displays velocity per rep as a simple chart after the set
- Detects the point of meaningful velocity loss (configurable threshold, default 20%)
- Alerts the user when velocity loss suggests stopping — before form breaks down

Velocity-Based Training Mode

A full VBT training mode for advanced users who want to train by velocity targets.

- Set target velocity ranges per exercise (e.g. strength: 0.3–0.5 m/s, power: 0.75–1.0 m/s)
- Real-time velocity feedback during the lift
- Auto-stop suggestion when target velocity zone is breached
- Velocity data logged and trends visible in Progress & Analytics
- Elite tier feature — requires V2.5 AI Upgrade Layer and wearable or camera input

Feature 33: AI Form Coach

Combines bar path and velocity data with pose estimation to give post-set form feedback. This is the full vision of Wali AI as a technical coach — not just a plan generator.

Post-Set Form Summary

After each set, the AI analyses the recorded movement and provides specific feedback.

- Identifies common technical breakdowns (e.g. forward lean, knee cave, bar drift)
- Prioritises the single most important cue rather than overwhelming the user
- References the user's history — 'Your squat depth has improved over the last 3 weeks'
- Distinguishes fatigue-related breakdown from technique issues

Form History & Trend Analysis

Tracks form quality over time so improvement is measurable.

- Form score per session per lift — logged automatically
- Visible in Progress & Analytics alongside strength trends
- AI flags when form is declining alongside increasing weight — injury risk signal
- Shareable form clips with squad for coaching and accountability

AI Architecture & Engine Strategy

waliFit does not build its own AI model — it builds an intelligent agent layer on top of commodity AI providers. The model is the engine. The agent design — system prompts, context injection, routing logic, and output schemas — is the competitive advantage.

> *Core principle: The AI model is a commodity. Any capable model can generate a training plan. What makes Wali AI feel like waliFit and not ChatGPT is the system prompt, the user context injected into every call, and the structured output schema that maps AI responses to your app's data model. This is your IP and nobody can see it.*

Model Selection: Gemini + Claude Hybrid

Using two providers with a model-agnostic abstraction layer gives waliFit the best balance of quality, cost, and resilience. The routing logic determines which model handles which task — the rest of the application never needs to know.

Task	Model	Why	Approx Cost
Program Architect (plan generation)	Claude Sonnet	Best structured reasoning for constrained plan generation	~$0.012/call
Daily Coach Chat (complex questions)	Claude Sonnet	Nuanced health & injury reasoning, careful tone	~$0.008/call

Daily Coach Chat (simple Q&A)	Gemini Flash	Fast, cheap, sufficient for basic questions	~$0.001/call
Workout adjustments	Gemini Flash	Low complexity, high volume — cost efficiency matters	~$0.001/call
Nutrition text parsing	Gemini Flash	Structured output, simple logic, very high volume	~$0.001/call
Food performance rating	Gemini Flash	Rule-based scoring with AI assist — cheap is right	~$0.001/call
Vision Meal Scanner	Gemini Vision	Best-in-class food image recognition	~$0.003/call
Program Evolution Engine	Gemini 1.5 Pro	Needs 1M token context for full training history	~$0.015/call
Bar Path & Velocity (V3)	Custom CV model	Real-time on-device inference — no API call	~$0/call

Agent Architecture: Three Layers

The agent wrapping the models is where the product intelligence lives. Three components must be engineered carefully before any AI feature ships.

Layer 1: System Prompt Engineering

The Wali AI system prompt is the product's brain — it defines everything the AI knows about its role, rules, and personality.

- Defines Wali AI's coaching philosophy and tone
- Sets hard rules — what it will and will not advise (injuries, extreme dieting, medical claims)
- Specifies structured output format for plan generation (JSON schema)
- Instructs the model on how to handle sensitive topics (pain, overtraining, mental health)
- This is your IP — invest weeks in getting it right before shipping
- Build a prompt testing harness: generate 50+ plans and ask 100+ edge-case questions before launch

Layer 2: Context Injection

Every API call silently includes the user's full relevant profile — the model has no memory between sessions so waliFit is the memory.

- User goals, experience level, equipment, and training schedule
- Current active training plan (full structure)
- Last 14 days of workout logs (exercise, sets, reps, RPE, notes)
- Nutrition averages over the last 7 days
- Current streak, tree state, and vitality score

– Any logged injuries or limitations
– Context is serialised into a standard template prepended to every call
– Context window budget managed carefully — trim older data first when approaching limits

Layer 3: Structured Output & Validation

For structured outputs like generated training plans, every response is validated against a schema before it reaches the UI.

– Training plan output schema: weeks → days → sessions → exercises → sets/reps/rest
– JSON schema validation on every plan generation response
– Fallback and retry logic if the model returns malformed output
– The app should never crash or show raw AI text because of unexpected model output
– Nutrition estimates returned as structured objects, not natural language paragraphs

Cost Estimates by Phase

Phase	Active Users	AI Calls/User/Month	Est. AI Cost/User	Total AI/Month
V1 Launch	0 – 1,000	~55 (all Claude)	~$0.18	< $180
V1 Growth	1,000 – 10,000	~55 (hybrid)	~$0.10	< $1,000
V2 (post-monetise)	10,000 – 50,000	~70 (more nutrition)	~$0.12	< $6,000
V2.5 Scale	50,000+	~80 (full stack)	~$0.14	< $7,000

> *At 50,000 paying users at $9.99/month, monthly revenue is ~$500,000. AI costs at that scale are under $7,000/month — less than 1.5% of revenue. AI cost is not a meaningful risk at any realistic early-stage scale.*

Permanently Deferred Features

These features are valuable but should not be scheduled until V2.5 is complete and user volume justifies the investment.

Feature	Reason for Delay
Skeletal lift analysis	Technically complex, requires hardware/computer vision infrastructure
Real-time form correction	High engineering cost, requires device-side ML or cloud vision pipeline

GPS ghost racer mode	Requires location infrastructure, niche use case for V1–V2 audience
Run club systems	Separate product scope — waliFit is strength + hybrid, not run-primary
Exercise video library	High content production cost — prioritize catalog over video
Biohacking claims	Requires scientific backing and legal review before any claims are made

Legal & Compliance Requirements

These are not optional. Missing any of them will either get the app rejected during App Store review or expose waliFit to legal risk post-launch. All must be completed before submitting to any app store.

Privacy Policy

Required by both Apple App Store and Google Play before submission. Must be a live, publicly accessible URL. Generate the base document using Termly (termly.io) or Iubenda (iubenda.com) — both are free and produce GDPR and CCPA compliant output. Then customize the following waliFit-specific sections.

What waliFit's Privacy Policy must specifically cover

Generic templates will miss these — they must be added manually.

– Health and fitness data: workouts, body weight, nutrition logs, hydration, injuries
– AI interaction data: that user messages and context are sent to Anthropic and Google AI for processing
– Behavioural signals collected for product improvement: plan acceptance, edits, completion rates
– Third parties named explicitly: Anthropic, Google (Gemini), Supabase, Firebase, Railway, Open Food Facts, Stripe
– User content: that messages and Commons posts are stored on waliFit servers
– Account deletion: exactly what data is deleted, how long it takes (state 30 days), and how to request it
– User rights: access, correction, deletion, and data portability rights under GDPR
– Age restriction: waliFit is not intended for users under 16
– Opt-out: users can opt out of AI training data collection without losing app functionality

Terms of Service

Required for App Store submission and essential liability protection. Generate the base at Termly or Iubenda, then add the following waliFit-specific clauses.

waliFit-specific Terms of Service clauses

These are your primary liability protections — do not skip them.

– AI disclaimer: Wali AI is a fitness guidance tool only. Nothing it says constitutes medical advice, diagnosis, or treatment. Users should consult a qualified health professional before starting any new exercise or nutrition programme.

– Age requirement: Users must be 16 or older. Users 16-18 require parental or guardian consent.

– User content: Posts and messages in waliFit grant waliFit a licence to display that content to other users. Users retain ownership.

– Community standards: waliFit reserves the right to remove content and suspend accounts that violate community guidelines.

– Subscription terms: Auto-renewal, cancellation policy, and refund eligibility must be stated exactly as required by Apple and Google.

– Service availability: waliFit does not guarantee uninterrupted service — this covers AI downtime and maintenance.

– Limitation of liability: waliFit is not liable for injury or health outcomes resulting from following AI-generated plans.

App Store Health App Compliance

Apple has specific review guidelines for health and fitness apps. Failure to comply results in rejection or removal post-launch.

Apple App Store — specific requirements for waliFit

Test all of these before submitting.

– Section 5.1.1 — Data Collection: App Review notes must explicitly state what health data is collected, why, and how it is protected. Write this in the App Review Notes field when submitting.

– Section 1.3 — Medical Claims: Wali AI must never say anything interpretable as a clinical diagnosis, treatment, or medical recommendation. The App Store review team will actively test this — run 50+ edge-case health questions through Wali AI before submission.

– Account deletion: Must be available from within the app. Apple will reject apps where users cannot delete their account in-app. (Covered by Settings screen — Feature 7.)

– Subscription handling: In-app purchases and subscriptions must use Apple's payment system exclusively on iOS — no linking to external payment pages.

– App Review test account: Provide a pre-seeded test account in App Review Notes with an active plan, logged workouts, and squad membership. Reviewers cannot evaluate the app on an empty account.

– HealthKit (V2.5): When wearable integration ships, request only the specific HealthKit permissions used. Broad HealthKit access requests without justification are grounds for rejection.

Google Play — specific requirements for waliFit

– Health and fitness app declaration: waliFit must be declared as a Health & Fitness app in the Play Console. This triggers additional data safety requirements.
– Data safety form: Must accurately disclose every type of data collected, whether it is shared with third parties, and whether users can request deletion. Inaccurate data safety forms result in removal.
– Account deletion: Google Play now requires that apps with account creation also provide a way for users to delete their account and data — both in-app and via a web form.
– Sensitive permissions: If microphone, camera, or location permissions are requested, the purpose must be clearly justified.

Pre-submission checklist — complete before any store submission

– Privacy Policy URL is live and publicly accessible
– Terms of Service URL is live and publicly accessible
– Account deletion flow is functional and tested
– Wali AI has been tested with 50+ edge-case health and injury questions
– AI disclaimer is visible in Settings and referenced in onboarding
– App Store Review Notes written with data collection explanation and test account credentials
– Data Safety form completed accurately in Play Console
– All subscription tiers tested end-to-end including cancellation flow
– App tested on minimum supported OS versions (iOS 16+, Android 10+)

Pricing & Monetization Strategy

waliFit launches completely free through V1 and V2 to build a user base and validate the product. Monetization is introduced at V2.5 through a tiered subscription model. The AI engine uses a Gemini + Claude hybrid approach to keep costs minimal while maintaining quality where it matters.

> *Core rule: Nothing that exists on free at launch ever gets taken away. The paywall applies to new premium features only. Users who built habits on the free tier should never feel punished for not upgrading.*

Tier 1 — waliFit Free

The full structural product. Genuinely useful on its own — not a crippled trial. Free exists to grow the community, fill squads, and drive word-of-mouth.

Included on Free	Not Available on Free

✓ Workout logging & custom builder ✓ Exercise library & plate calculator ✓ Rest timer & offline mode ✓ Calendar (full — daily, weekly, monthly) ✓ Vitality Tree (full — all growth states) ✓ Streaks & consistency tracking ✓ Squads, PR feed, friends & messaging ✓ Smart notifications ✓ Barcode scanner (basic nutrition lookup) ✓ Body weight logging	– Wali AI — all features – Nutrition AI (meal text parsing, food rating) – Progress & Analytics (charts, PR vault) – Body measurements & progress photos – Home screen widget – Global Arena & leaderboards – Vision Meal Scanner – Wearable integration & HRV coaching – AI program evolution engine – Reward unlocks & Nexus Seed economy

> *Vitality Tree stays fully intact on free. It is the emotional core of the product and the primary word-of-mouth driver. Capping or degrading the tree on free would damage the most valuable retention mechanic before monetization even begins.*

Tier 2 — waliFit Pro

waliFit Pro **$9.99 / month** $79.99 / year (save 33%) *For the everyday hybrid athlete who wants AI coaching and performance tracking.*	**Everything in Free, plus:** ✓ Wali AI — Program Architect, Daily Coach Chat, Workout Adjustment ✓ Nutrition AI — meal text parsing, food performance rating, ingredient analysis ✓ Progress & Analytics — strength trends, frequency charts ✓ Personal PR Vault ✓ Body measurements & progress photos ✓ Home screen widget ✓ Reward unlocks & Nexus Seed economy ✓ Priority support

Tier 3 — waliFit Elite

waliFit Elite **$14.99 / month** $119.99 / year (save 33%) *For competitive hybrid athletes who want the full system — AI, competition, and wearable integration.*	**Everything in Pro, plus:** ✓ Global Arena — Workout of the Week & Month ✓ Global, local & social leaderboards ✓ RX & Scaled competition divisions ✓ Vision Meal Scanner ✓ Energy & bloat tracking

	✓ AI Program Evolution Engine ✓ Recovery & deload suggestions ✓ Improved Daily Coach memory ✓ Wearable integration (Apple Watch, Garmin) ✓ HRV-driven coaching adjustments ✓ Early access to all future premium features

Pricing Rationale

Why these price points

waliFit sits between a basic tracker and a full AI coaching service.

- Hevy (workout tracker): $9.99/month — waliFit Pro matches this and delivers significantly more
- Strava (social fitness): $11.99/month — waliFit Pro undercuts it with a stronger feature set
- TrainHeroic (programming): ~$13/month — waliFit Pro is competitive and more social
- AI coaching apps (e.g. Future): $150+/month — waliFit Elite at $14.99 is a compelling AI alternative
- The $5 gap between Pro and Elite is intentional — low friction to upgrade, high value difference

Annual Plan Strategy

Annual plans are the most important conversion target.

- 33% discount on annual vs monthly — standard across the fitness app market
- Annual subscribers churn at significantly lower rates than monthly
- Improved cashflow — revenue collected upfront rather than month-to-month
- Consider a launch promotion: first year 40% off to accelerate early annual conversions

Conversion Strategy

How to move free users to paid.

- AI features are locked but visible on free — users see what they are missing
- After logging 3+ workouts, prompt a soft upsell: 'Let Wali AI build your next training block'
- After a PR, prompt: 'Track your full progress history with waliFit Pro'
- Squad challenges in V2 only available to Pro/Elite — social pressure drives upgrades
- Never use countdown timers, dark patterns, or aggressive popups — the product should sell itself

Roadmap Summary

Release	Focus	Key Features	Goal
V1	Full Launch	Wali AI, Workout Engine, Vitality Tree, WaliRun (GPS + Run Stats), The Arena (Workout Squads + Run Clubs), Nutrition Logger, Settings	Ship legally compliant — tree fully functional, settings complete, cold start solved
V2	Intelligence Depth	Smart Fuel, Tree Biomes, Squad Forests, RPE Timer, Analytics, Body Metrics, Dashboard, waliFit Commons, Referrals	Deepen the experience and introduce monetization-ready features
V2.5	Competition + Premium	Global Arena, Adaptive AI, Station-to-Run Timer, Vision Meal Scanner, Wearables	Scale and introduce paid tiers
V3	Mechatronics Layer	Bar Path Tracking, Velocity Loss Monitoring, AI Form Coach	Build the long-term technical moat using mechatronics expertise